

Problem

On average, how many times must a fair die be thrown until the first 6 appears?

Approach

Let p be the probability of rolling a 6 on any throw, and let $q = 1 - p$.

The probability that the first success occurs on the k -th trial is

$$P(X = k) = pq^{k-1}, \quad k = 1, 2, \dots$$

Use the definition of expectation to compute the mean number of trials.

Solution

The expected number of throws is

$$E[X] = 1 \cdot p + 2 \cdot pq + 3 \cdot pq^2 + \dots$$

Let

$$m = E[X].$$

Multiplying by q ,

$$qm = pq + 2pq^2 + 3pq^3 + \dots$$

Subtracting,

$$\begin{aligned} m - qm &= p + pq + pq^2 + \dots \\ &= p(1 + q + q^2 + \dots) \\ &= \frac{p}{1 - q}. \end{aligned}$$

Since $q = 1 - p$,

$$m(1 - q) = 1,$$

which gives

$$mp = 1.$$

Therefore,

$$m = \frac{1}{p}.$$

For a fair die,

$$p = \frac{1}{6},$$

so

$$\boxed{E[X] = 6.}$$

Key Insight

The number of trials until the first success follows a geometric distribution, whose expected value is simply

$$\boxed{\frac{1}{p}}.$$

For a fair die, this gives an average of six throws to obtain the first 6.